

COMPONENTS

Schottky barrier rectifiers

Littelfuse, Inc. has introduced DST series of Schottky barrier rectifiers designed to outperform conventional switching diodes in commercial, industrial and automotive applications. These rectifiers combine ultra-low forward voltage drop, high current handling, high junction temperature capability and low leakage performance in a compact TO-277B surface-mount package. For automotive applications up to 10A, these are said to offer the same or better performance as Schottky barrier rectifiers provided in D-PAK packaging, with a footprint that's only one-third of the size.

Littelfuse, Inc.
www.littelfuse.com

Hybrid power and signal connector

CONEC has launched a hybrid power and signal circular connector that saves time and costs for cabling, installation and maintenance for



conventional wiring of applications like motors, servo drives, linear units/drives and more. By combining power supply and data transmission into one single connector, the compact SuperCon Series hybrid connector system saves space by eliminating the need for an additional wire and connector. The rugged IP67-rated connector features a safe and secure connection via a bayonet locking mechanism.

CONEC
www.conec.com/en/conec-supercon-connectors/

WiFi + BLE SoC

Uchi Embedded Solutions has introduced ESP32 Wi-Fi (2.4GHz) and Bluetooth combo chip from Espressif Systems that uses TSMC's ultra-low-power 40nm technology. The chip is designed for mobile, wearable electronics and the Internet of Things (IoT) applications, with features like fine-resolution clock gating, power modes and dynamic power scaling.

For instance, in a low-power IoT sensor hub application scenario, ESP32 is woken up periodically and only when a specified condition is detected; low duty cycle is used to minimise the amount of energy that the chip expends. The output power of the power amplifier is adjustable to achieve an optimal trade-off between communication range, data rate and power consumption.

Uchi Embedded Solutions
www.uchiembedded.co.in

SoC for vehicle infotainment

Renesas Electronics has started volume shipment of its first R-Car system-on-chip (SoC) incorporating Automotive Grade Linux (AGL)-based software. AGL is a cross-industry effort to develop a fully open software stack for the connected car. The R-Car starter kit and AGL-based software accelerate the development of automotive in-vehicle infotainment (IVI) applications, and move automakers closer to the realisation of connected cars.

Renesas
www.renesas.com

Transceiver

The MAX14827A from Maxim Integrated is a tiny input/output-link device transceiver. It integrates high-

voltage functions commonly found in industrial sensors, including drivers and regulators.

The device features two ultra-low-power drivers with active reverse-polarity protection. Operation is specified for normal 24V supply voltages up to 60V. Pin control logic inputs allow operation with switching sensors that do not use a microcontroller. For sensors that use a microcontroller, an SPI interface is available with extensive diagnostics.

Maxim Integrated
www.maximintegrated.com

Single-die LED

Lumileds has revealed its highest flux density LED yet, the LUXEON V. Featuring a compact 4x4mm² footprint, the new LED produces more than 1700 lumens at 5700K and 70CRI (4.8A)—50 per cent higher output than competing LEDs. It is said to enable fixture designs that were previously unattainable for stadium and area lighting, torch and other portables because of the unique combination of high efficacy at high drive current with extremely low thermal resistance and optimised radiation pattern.

Lumileds
www.lumileds.com/LUXEONV

TEST & MEASUREMENT

Digital multimeter

Kyoritsu's 1021R digital multimeter can measure current up to 10A. Its key features include accurate reading with true RMS, a large display with 6000 counts and backlight, 600V AC/DC auto ranging, 0.1mV resolution, frequency

